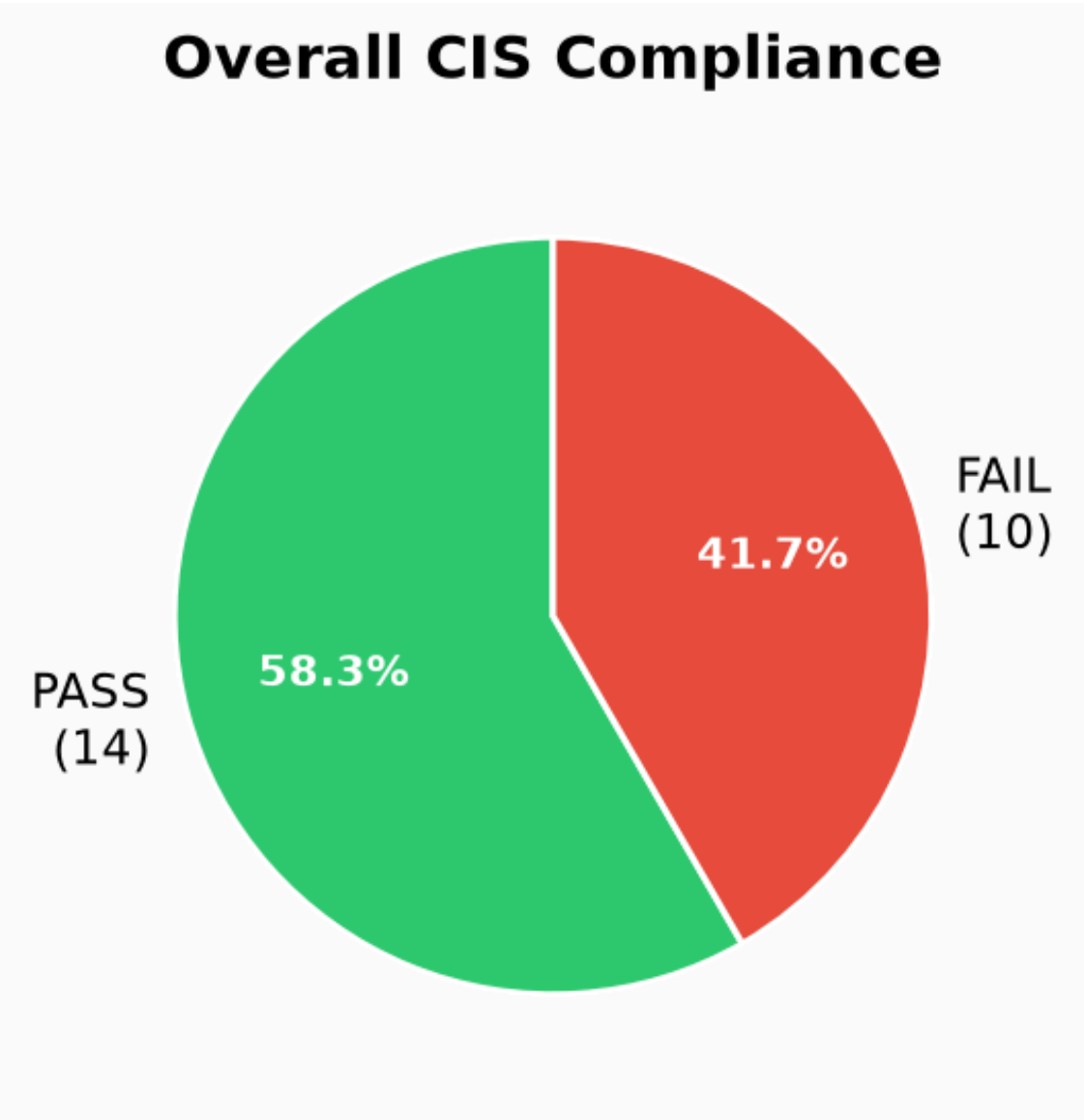


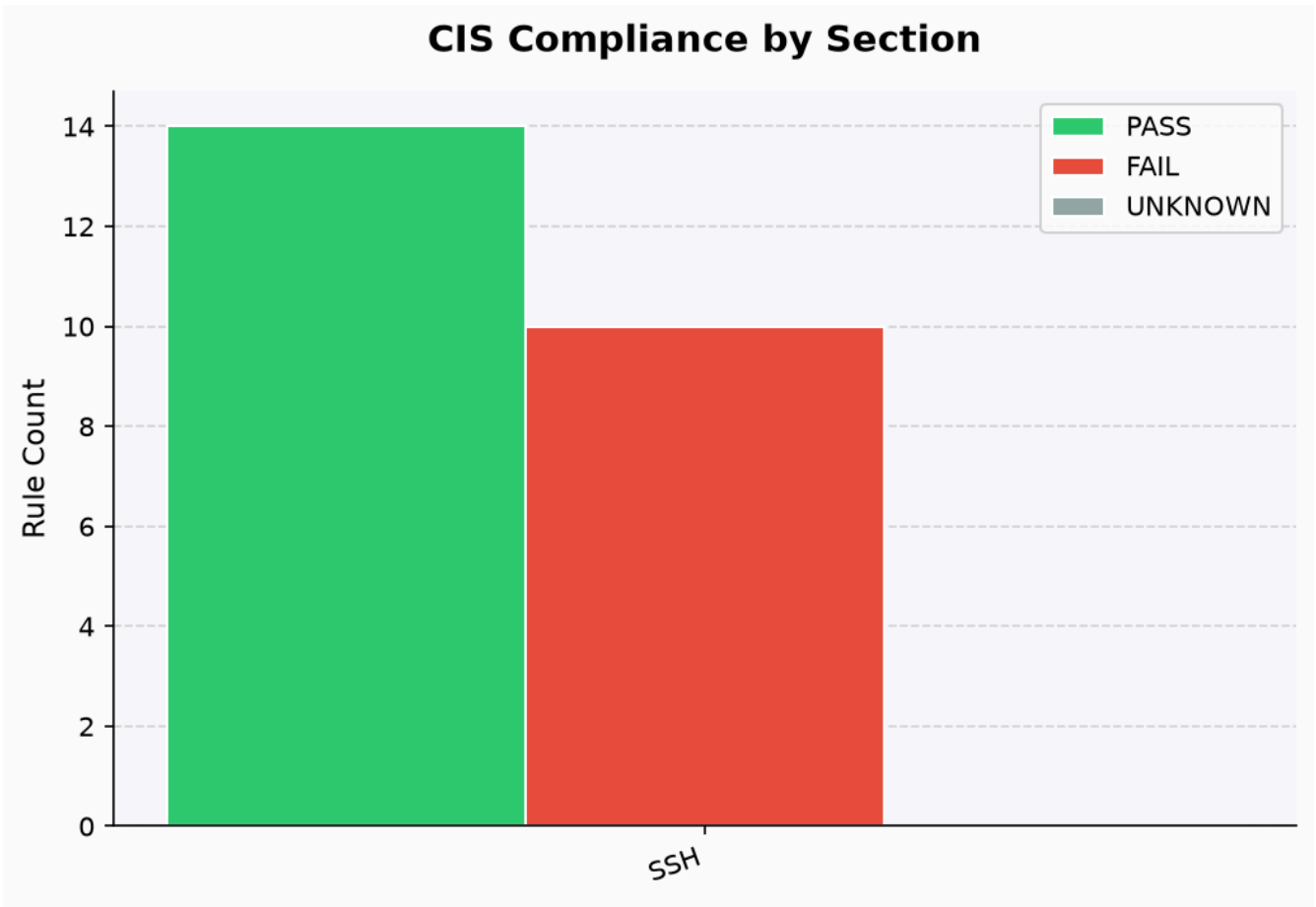
SENTINEL — CIS Benchmark Audit Report

Executive Summary

Metric	Value
Total Rules Evaluated	24
✓ Passed	14
✗ Failed	10
? Unknown	0
Overall Pass Rate	58.3%

Compliance Charts





Per-Section Breakdown

Section	✓ PASS	✗ FAIL	? UNKNOWN	Pass Rate
SSH	14	10	0	58%

CIS Benchmark Compliance Report

SSH Rules

Rule 5.1.1: Ensure access to /etc/ssh/sshd_config is configured (Automated)

- **Status:** PASS
- **Evidence:** Permissions for /etc/ssh/sshd_config are 644 root root.
- **Recommendation:** None

Rule 5.1.2: Ensure access to SSH private host key files is configured (Automated)

- **Status:** PASS
- **Evidence:** Permissions for SSH private host key files are 600 root root for /etc/ssh/ssh_host_ecdsa_key, /etc/ssh/ssh_host_ed25519_key, and /etc/ssh/ssh_host_rsa_key.

- **Recommendation:** None

Rule 5.1.3: Ensure access to SSH public host key files is configured (Automated)

- **Status:** PASS
- **Evidence:** Permissions for SSH public host key files are 644 root root for /etc/ssh/ssh_host_ecdsa_key.pub, /etc/ssh/ssh_host_ed25519_key.pub, and /etc/ssh/ssh_host_rsa_key.pub.
- **Recommendation:** None

Rule 5.1.4: Ensure sshd access is configured (Automated)

- **Status:** FAIL
- **Evidence:** ListenAddress is configured to 0.0.0.0:22, which allows SSH to listen on all available network interfaces.
- **Recommendation:** Configure ListenAddress in /etc/ssh/sshd_config to a specific IP address to restrict SSH access to a designated interface. For example, ListenAddress 192.168.1.100:22.

Rule 5.1.5: Ensure sshd Banner is configured (Automated)

- **Status:** FAIL
- **Evidence:** Banner is set to none.
- **Recommendation:** Configure Banner in /etc/ssh/sshd_config to point to a warning banner file (e.g., Banner /etc/issue.net) to display a warning message before authentication.

Rule 5.1.6: Ensure sshd Ciphers are configured (Automated)

- **Status:** PASS
- **Evidence:** Configured ciphers are chacha20-poly1305@openssh.com, aes128-ctr, aes192-ctr, aes256-ctr, aes128-gcm@openssh.com, aes256-gcm@openssh.com. These are considered strong ciphers.
- **Recommendation:** None

Rule 5.1.7: Ensure sshd ClientAliveInterval and ClientAliveCountMax are configured (Automated)

- **Status:** FAIL
- **Evidence:** ClientAliveInterval is set to 0, which disables the feature.
- **Recommendation:** Configure ClientAliveInterval to a non-zero value (e.g., 300 seconds) and ClientAliveCountMax (e.g., 3) in /etc/ssh/sshd_config to terminate idle SSH sessions.

Rule 5.1.8: Ensure sshd DisableForwarding is enabled (Automated)

- **Status:** FAIL
- **Evidence:** AllowTcpForwarding is true, AllowAgentForwarding is true, AllowStreamLocalForwarding is true, PermitOpen is any, and PermitListen is any.
- **Recommendation:** Disable all forms of forwarding by setting AllowTcpForwarding

no, AllowAgentForwarding no, AllowStreamLocalForwarding no,
PermitOpen none, and PermitListen none in /etc/ssh/sshd_config.

Rule 5.1.9: Ensure sshd GSSAPIAuthentication is disabled (Automated)

- **Status:** PASS
- **Evidence:** GSSAPIAuthentication is set to false.
- **Recommendation:** None

Rule 5.1.10: Ensure sshd HostbasedAuthentication is disabled (Automated)

- **Status:** PASS
- **Evidence:** HostbasedAuthentication is set to false.
- **Recommendation:** None

Rule 5.1.11: Ensure sshd IgnoreRhosts is enabled (Automated)

- **Status:** PASS
- **Evidence:** IgnoreRhosts is set to true.
- **Recommendation:** None

Rule 5.1.12: Ensure sshd KexAlgorithms is configured (Automated)

- **Status:** FAIL
- **Evidence:** The system is configured with weak Key Exchange Algorithms including ecdh-sha2-nistp256, ecdh-sha2-nistp384, ecdh-sha2-nistp521 (suspected NIST/NSA backdoor concerns), and diffie-hellman-group14-sha256 (deprecated or known weaknesses).
- **Recommendation:** Remove weak and deprecated KexAlgorithms from the KexAlgorithms directive in /etc/ssh/sshd_config. Prioritize modern, strong, and post-quantum safe algorithms.

Rule 5.1.13: Ensure sshd LoginGraceTime is configured (Automated)

- **Status:** FAIL
- **Evidence:** LoginGraceTime is set to 120 seconds.
- **Recommendation:** Configure LoginGraceTime in /etc/ssh/sshd_config to a value between 30 and 60 seconds (e.g., 60) to reduce the window for brute-force attacks.

Rule 5.1.14: Ensure sshd LogLevel is configured (Automated)

- **Status:** PASS
- **Evidence:** LogLevel is set to INFO.
- **Recommendation:** None

Rule 5.1.15: Ensure sshd MACs are configured (Automated)

- **Status:** FAIL
- **Evidence:** The system is configured with weak MAC algorithms including umac-64-etm@openssh.com, hmac-sha1-etm@openssh.com (enables Terrapin attack),

umac-64@openssh.com, umac-128@openssh.com, hmac-sha2-256, hmac-sha2-512, and hmac-sha1 (SHA-1 hash, collision attacks demonstrated).

- **Recommendation:** Remove weak and deprecated MAC algorithms from the MACs directive in `/etc/ssh/sshd_config`. Prioritize modern, strong MACs like `hmac-sha2-512-etm@openssh.com` or `hmac-sha2-256-etm@openssh.com`.

Rule 5.1.16: Ensure sshd MaxAuthTries is configured (Automated)

- **Status:** PASS
- **Evidence:** `MaxAuthTries` is set to 6.
- **Recommendation:** None

Rule 5.1.17: Ensure sshd MaxStartups is configured (Automated)

- **Status:** PASS
- **Evidence:** `MaxStartups` is configured to 10:30:100.
- **Recommendation:** None

Rule 5.1.18: Ensure sshd MaxSessions is configured (Automated)

- **Status:** PASS
- **Evidence:** `MaxSessions` is set to 10.
- **Recommendation:** None

Rule 5.1.19: Ensure sshd PermitEmptyPasswords is disabled (Automated)

- **Status:** PASS
- **Evidence:** `PermitEmptyPasswords` is set to false.
- **Recommendation:** None

Rule 5.1.20: Ensure sshd PermitRootLogin is disabled (Automated)

- **Status:** FAIL
- **Evidence:** `PermitRootLogin` is set to `without-password`.
- **Recommendation:** Configure `PermitRootLogin no` in `/etc/ssh/sshd_config` to prevent direct root login via SSH. Users should log in with their own accounts and use `sudo` for privilege escalation.

Rule 5.1.21: Ensure sshd PermitUserEnvironment is disabled (Automated)

- **Status:** PASS
- **Evidence:** `PermitUserEnvironment` is set to false.
- **Recommendation:** None

Rule 5.1.22: Ensure sshd UsePAM is enabled (Automated)

- **Status:** PASS
- **Evidence:** `UsePAM` is set to true.
- **Recommendation:** None

Rule 5.1.23: Ensure sshd post-quantum cryptography key exchange algorithms are configured (Automated)

- **Status:** FAIL
- **Evidence:** The system is configured with weak Key Exchange Algorithms including `ecdh-sha2-nistp256`, `ecdh-sha2-nistp384`, `ecdh-sha2-nistp521`, and `diffie-hellman-group14-sha256`, which are not post-quantum cryptography (PQC) algorithms and some have known weaknesses.
- **Recommendation:** Configure `KexAlgorithms` in `/etc/ssh/sshd_config` to include and prioritize post-quantum cryptography key exchange algorithms, and remove all weak or non-PQC algorithms.

Rule 5.1.24: Ensure sshd ListenAddress is configured (Automated)

- **Status:** FAIL
 - **Evidence:** `ListenAddress` is configured to `0.0.0.0:22`, which allows SSH to listen on all available network interfaces.
 - **Recommendation:** Configure `ListenAddress` in `/etc/ssh/sshd_config` to a specific IP address to restrict SSH access to a designated interface. For example, `ListenAddress 192.168.1.100:22`.
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